

MARC LICHTMAN

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EDUCATION

Ph.D., Electrical Engineering, Expected 2015

Virginia Tech, Blacksburg, VA

- Areas of Interest: Electronic Warfare, Machine Learning, LTE and LTE Advanced
- Dissertation Topic: Robust Satellite Communications under Hostile Interference

M.S., Electrical Engineering, Obtained December 2012

Virginia Tech, Blacksburg, VA

B.S., Electrical Engineering, Obtained May 2011

Virginia Tech, Blacksburg, VA

- Graduated Magna Cum Laude with a GPA of 3.7 on a 4.0 scale

PAST POSITIONS

Instructor - Virginia Tech

Summer 2013

- Taught the undergraduate course *ECE 2204: Electronics I*
- Topics included semiconductor physics, digital circuits, and DC analysis of diodes, BJTs, and MOFETs

Graduate Research Assistant - Virginia Tech

January 2012 - Present

- Currently investigating methods of enabling robust satellite communications under hostile interference
- Analyzed the vulnerability of LTE to hostile and sophisticated cross-layer jamming
- Surveyed the current state of Dynamic Spectrum Access (DSA) and LTE

Graduate Teaching Assistant - Virginia Tech

Fall 2011

- Instructed three undergraduate electronic circuits laboratories

Engineering Graduate Intern - GE Energy

Summer 2011

- Worked on the control system of GE's 700 kW solar inverter & 1.5 MW wind turbine

Engineering Intern - Overlook Systems Technologies

Summer 2010

- Designed and developed software models to simulate navigation warfare scenarios
- Analyzed methods of GPS jamming and spoofing

Engineering Co-op - Lutron Electronics

Summer and Fall 2009

- Designed and built an embedded system to be used as an engineering tool
- Redesigned an existing product, including analog circuit analysis, preliminary testing, worst-case analysis, KEMA certification testing and implementation
- Collaborated with corporate quality to design a "lessons learned" database
- Helped implement a new idea for use in a real time illumination stability system

Undergraduate Researcher - Virginia Tech

Summer 2008

- Utilized Linux, programming networking scripts in C
- Obtained experience working with multiple professors in a research environment

PUBLICATIONS

- 1) M. Lichtman, J. H. Reed, T. C. Clancy, M. Norton, "Vulnerability of LTE to Hostile Interference," *IEEE Global Conference on Signal and Information Processing*, December 2013. [Invited, Awaiting Publication]
- 2) M. Lichtman, W. C. Headley, J. H. Reed, "Automatic Modulation Classification under IQ Imbalance using Supervised Learning," *IEEE Military Communications Conference (MILCOM)*, November 2013. [Accepted, Awaiting Publication]
- 3) T. C. Clancy, M. Norton, M. Lichtman, "Security Challenges with LTE-Advanced Systems and Military Spectrum," *IEEE Military Communications Conference (MILCOM)*, November 2013. [Accepted, Awaiting Publication]
- 4) C. Shahriar, M. La Pan, M. Lichtman, T. C. Clancy, R. McGwier, R. Tandon, S. Sodagari, J. H. Reed, "OFDM Jamming Tutorial Paper," *Communications Surveys & Tutorials, IEEE* [Under Review]
- 5) S. Dudley, W. C. Headley, M. Lichtman, E. Imana, X. Ma, M. Abdelbar, A. Padaki, A. Ullah, M. Sohul, T. Yang, J. H. Reed, "Practical Issues for Spectrum Management with Cognitive Radios," *Proceedings of the IEEE*, March 2014 [Invited, Awaiting Publication]
- 6) M. La Pan, M. Lichtman, T. C. Clancy, R. W. McGwier, "Protecting Physical Layer Synchronization: Mitigating Attacks against OFDM Acquisition," *Global Wireless Summit 2013*, June 2013
- 7) J. H. Reed and M. Lichtman, "Response to FirstNet NOI regarding the Development of the Nationwide Interoperable Public Safety Broadband Network," November 2012. Available: http://www.ntia.doc.gov/files/ntia/va_tech_response.pdf

AWARDS & ACHIEVEMENTS

- Research featured in an MIT Technology Review article
- Winner of the WICAT Board Meeting Student Poster Contest
- Intelligence Community Center for Academic Excellence (IC-CAE) Scholarship
- Virginia Tech College of Engineering Fellowship
- Pamplin Leader Scholarship
- Bernard Silverman Scholarship
- Winner of the LabVIEW Invitational Programming Contest